Estimating Implicit Regularization in Deep Learning

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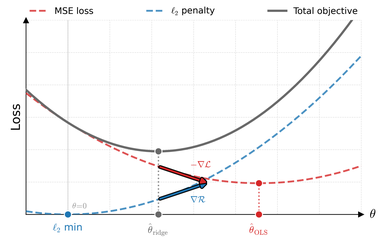
What regularization is implicit in your training procedure?

Why regularization? Explicit regularization adds penalties to the objective function that steer learning toward certain solutions—for example, those with smaller weights—in ways that can reduce overfitting and improve generalization. Training choices such as early stopping, minibatching, and dropout also bias which solutions are learned, but we often don't know exactly how [1, 2].

Characterizing implicit regularization. Theory characterizes some of these implicit effects, mostly in simplified settings [3, 4]. We propose an empirical framework for testing theoretical predictions and extending the study of implicit regularization to networks that are not analytically tractable.

Our approach. Work backwards: observe the learned weights and training dynamics, and estimate a parameterized regularizer that explains them. We achieve this by matching loss gradients or update residuals to the gradients of a proposed regularizer, then solving for unknown parameters.

Gradient matching



Estimating from learned weights

At a hypothetical minimum of $\mathcal{L} + \mathcal{R}$, the gradients counteract one another:

$$-\nabla_\theta \mathcal{L}(\hat{\theta}, D) = \nabla_\theta \mathcal{R}(\hat{\theta}, D; \Lambda).$$

Fit the regularizer parameters with the weights held fixed:

$$\hat{\Lambda} = \arg\min_{\Lambda} \frac{1}{p} \left\| \nabla_\theta \mathcal{L} + \nabla_\theta \mathcal{R}_\Lambda \right\|_2^2.$$

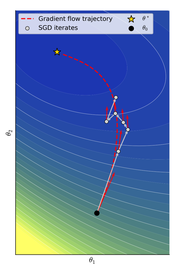
Estimating from weight updates

SGD iterates deviate from the full-batch gradient-flow path (right). For a gradient-step reference, match the update residual:

$$-\frac{\Delta\theta_t}{\eta} - \nabla_\theta \mathcal{L}(\theta_t) \;\approx\; \nabla_\theta \mathcal{R}(\theta_t; \Lambda),$$

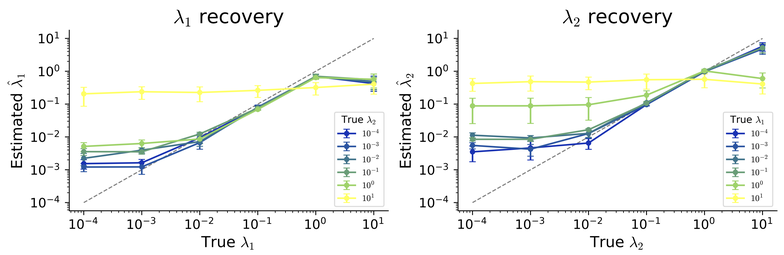
where $\Delta\theta_t = \theta_{t+1} - \theta_t$.

Stack steps to estimate a shared regularizer, or fit time-varying parameters. Setting the update to zero recovers the endpoint perspective.



Recovering known regularization

Recovering elastic-net coefficients



Deep ReLU networks; 6×6 penalty grid, 10 datasets per cell. Mean ± SE; dashed line = exact recovery.

Recovery is accurate at intermediate strengths. Weak signals and nearly collinear gradients limit identification at the extremes.

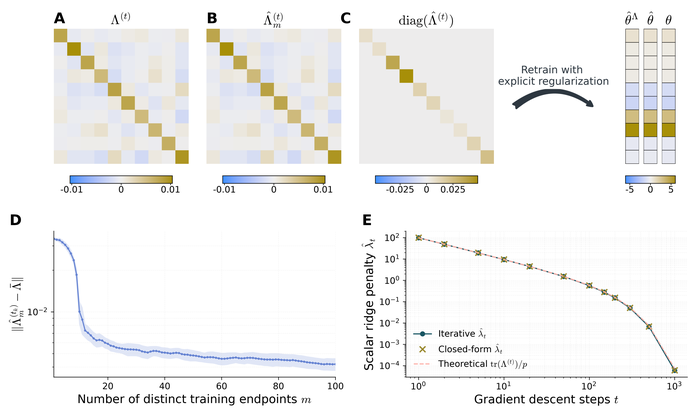
Early stopping as a quadratic penalty

The early-stopped OLS iterate solves a regularized problem [5]:

$$\hat{\theta}_t = \arg\min_\theta \left\{ \frac{1}{n}\|y - X\theta\|_2^2 + \mathcal{R}_t(\theta) \right\},$$

$$\mathcal{R}_t(\theta) = \theta^\top \Lambda^{(t)} \theta, \qquad \Lambda^{(t)} = VS\left[(I - \eta S)^{-t} - I\right]^{-1} V^\top.$$

Here $X^\top X/n = VSV^\top$, $\theta_0 = 0$, and $0 < \eta s_i < 1$. The step convention is $\theta_{t+1} = \theta_t + \eta X^\top (y - X\theta_t)/n$.

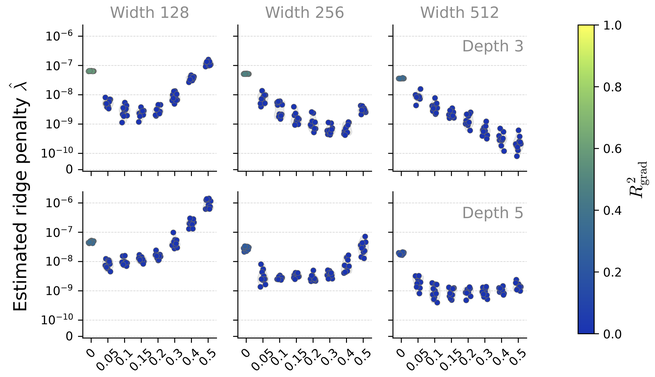


A–B: recover the theoretical matrix. **C:** a diagonal fit reproduces weights. **D–E:** more endpoints improve estimation; the penalty decays over training.

Implicit regularization in deep networks

Dropout rate and network architecture

Dropout has data-dependent regularization effects [6]. On MNIST, vary depth, width, and dropout rate; fit $\mathcal{R}(\theta, \lambda) = \lambda\|\theta\|_2^2$.

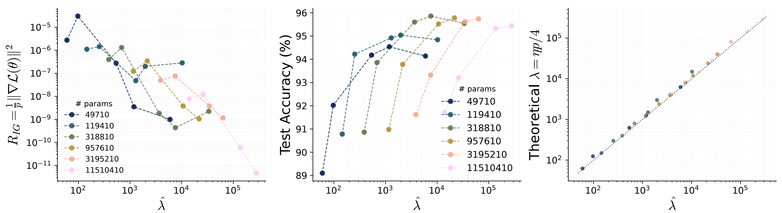


Each point is one of 10 seeds. Color shows gradient fit: lighter = more explained. Zero-dropout controls are included.

Narrow networks show an upturn at high dropout; wider networks have smaller estimates. Scalar ridge summarizes part of the effect.

Implicit gradient regularization

Barrett & Dherin predict $\mathcal{R} = \frac{\lambda}{p}\|\nabla\mathcal{L}\|^2$ with $\lambda = \eta p/4$ [7]. Estimates from deviations of discrete GD from continuous gradient flow track theory (right).



Contributions

- **Estimate implicit bias** by matching loss gradients to a proposed regularizer.
- **Characterize success:** recover known regularization or reproduce the original model's weights.
- **Bridge theory and practice:** validate analytical results and diagnose bias in complex networks.

References

[1] Peter L. Bartlett, Andrea Montanari, and Alexander Rakhlin. Deep learning: a statistical viewpoint. *Acta Numerica*, 2021.

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[4] Sanjeev Arora, Nadav Cohen, Wei Hu, and Yuping Luo. Implicit regularization in deep matrix factorization, 2019. arXiv:1905.13655.

[5] Alnur Ali, J. Zico Kolter, and Ryan J. Tibshirani. A continuous-time view of early stopping for least squares regression. In *AISTATS*, 2019.

[6] Colin Wei, Sham Kakade, and Tengyu Ma. The implicit and explicit regularization effects of dropout. In *ICML*, 2020.

[7] David G. T. Barrett and Benoit Dherin. Implicit gradient regularization, 2022. arXiv:2009.11162.

Paper: arXiv:2605.05436 **Blog:** jrudoler.com/estimating-implicit-regularization/ **Code:** github.com/jrudoler/estimating-implicit-regularization

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